

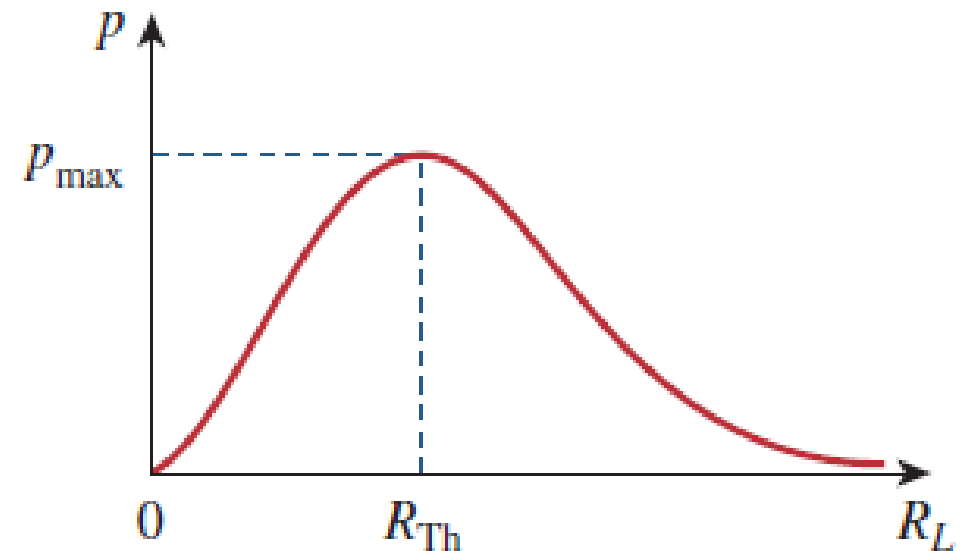
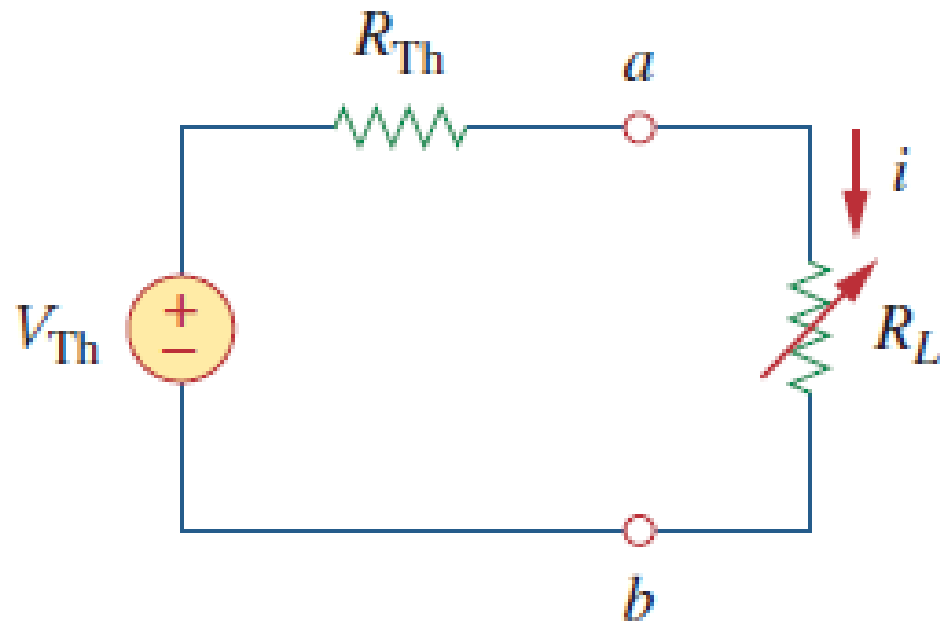
Module -1

DC Circuit Analysis and Theorems

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Maximum Power Transfer Theorem

Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).



Maximum Power Transfer Theorem

$$R_L \neq R_{Th}$$

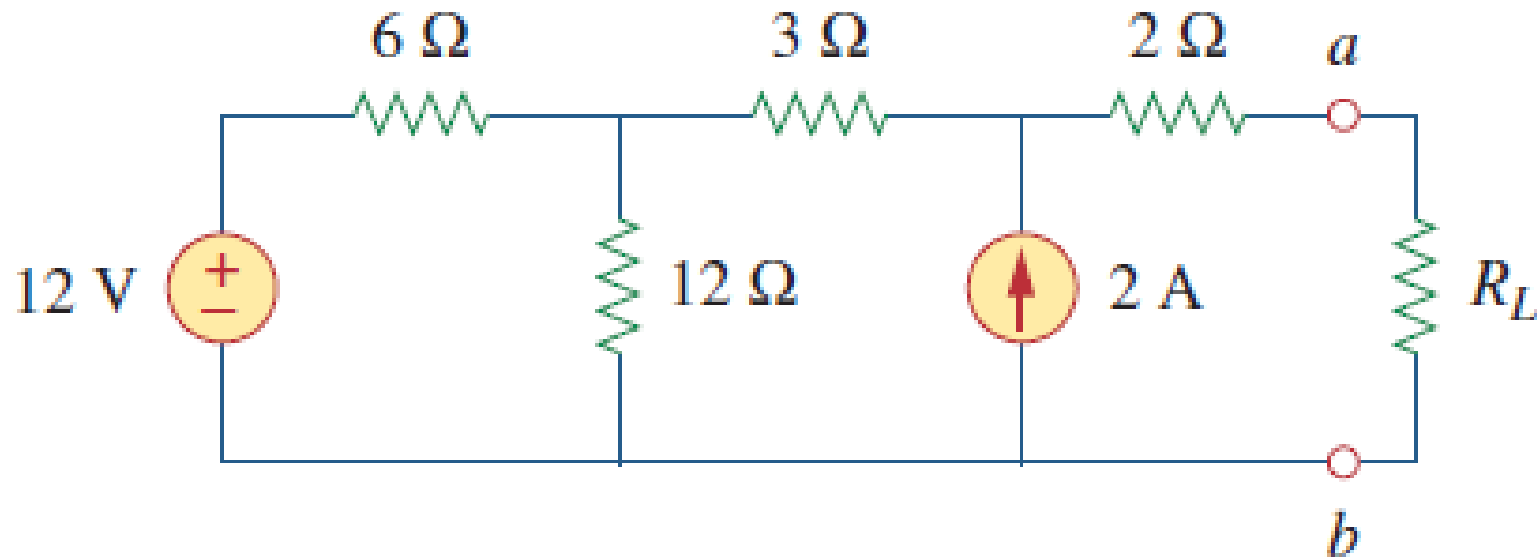
$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

$$R_L = R_{Th}$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

Tutorial - 1

Find the value of R_L for maximum power transfer in the circuit. Find the maximum power.



Solution:

$$R_{Th} = 2 + 3 + 6 \parallel 12 = 5 + \frac{6 \times 12}{18} = 9 \Omega$$

$$-12 + 18i_1 - 12i_2 = 0, \quad i_2 = -2 \text{ A}$$

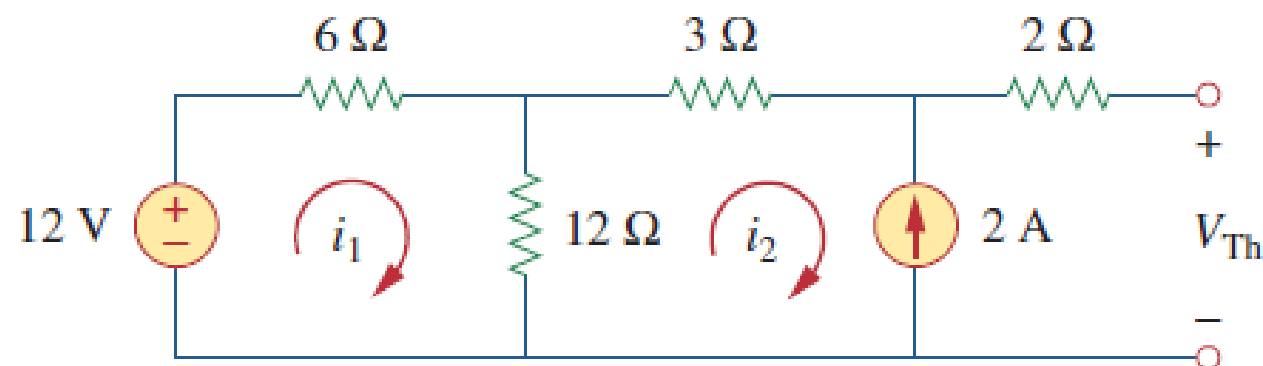
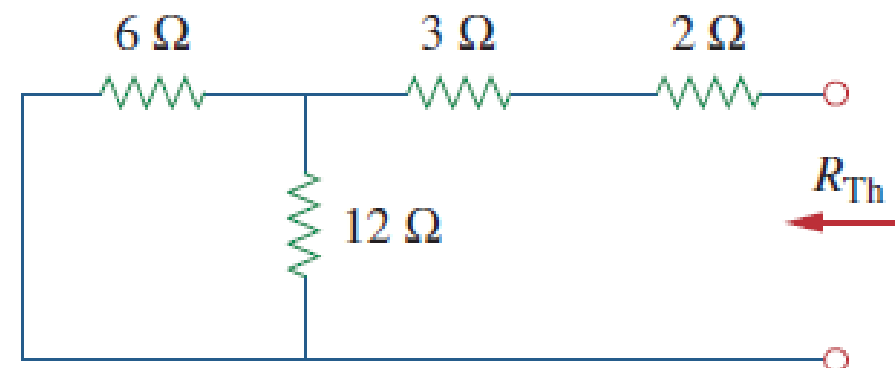
$$-12 + 6i_1 + 3i_2 + 2(0) + V_{Th} = 0$$

$$V_{Th} = 22 \text{ V}$$

For maximum power transfer,

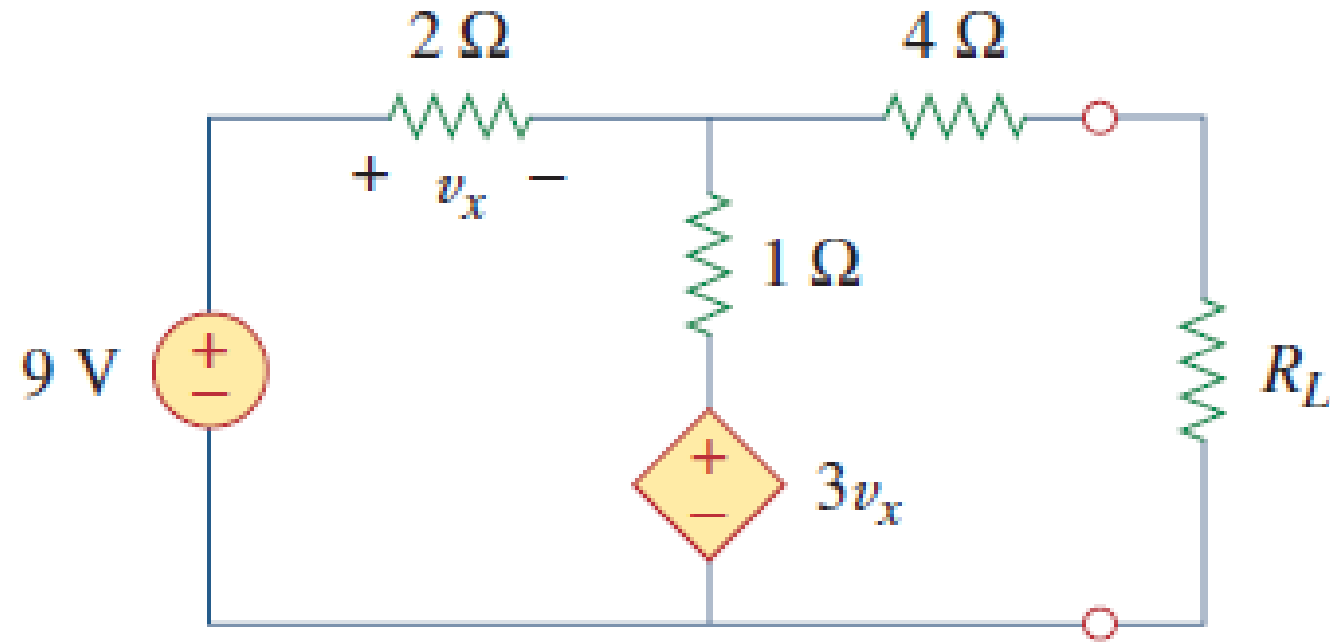
$$R_L = R_{Th} = 9 \Omega$$

$$p_{\max} = \frac{V_{Th}^2}{4R_L} = \frac{22^2}{4 \times 9} = 13.44 \text{ W}$$



Tutorial - 2

Determine the value of R_L that will draw the maximum power from the rest of the circuit. Calculate the maximum power.



Answer: $4.222\ \Omega$, $2.901\ \text{W}$.